Certified Adaptive Examination of Peripheral Blood Smears: When Information-Gain Stopping Fails, and What Can Be Guaranteed Instead

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***Abstract*—Automated examination of blood smears increasingly decides which fields to inspect and when to stop. We ask what it can guarantee about unexamined material, using pre-registered simulations from 28,280 expert-labelled cells, two external datasets and annotated bone-marrow regions from 99 patients. First, information-saturation stopping, reimplemented from InfoDPP-PAC, stopped at nearly the same point at every tested level of decision-changing evidence; with one critical field among 200 it missed 76% of positive specimens (uniform sampling of equal size: 85%), and on real marrow tiles it did worse than uniform sampling in low-blast specimens. Second, without location information, any strategy must examine nearly every field (95% at a 5% error level) to certify that rare evidence is absent. Third, certifying that the blast proportion exceeds a 10% or 20% threshold is feasible with design-based, anytime-valid tests whose cost in verified cells is governed mainly by the margin to the threshold. Near the threshold, a 100-cell count was wrong in up to 37% of specimens; valid procedures were almost never wrong but mostly undetermined. Fourth, validity held when acquisition shifts cut classifier blast recall from 0.91 to 0.34, while the saving from classifier-driven sampling reversed. Validity depended on the unit of analysis and the verifier: analysing cells of sampled fields as independent draws exceeded the nominal error in 27 of 75 real specimens, and a single reader as verifier yielded certificates about that reader's count. A classifier can reduce verification effort; validity depends on the design, the unit of analysis and weighting, and verification.**

***Index Terms*—Adaptive sampling, anytime-valid inference, blast count, blood smear, risk-limiting audit, sequential testing.**

I. Introduction

A peripheral blood smear contains far more cells than any examiner inspects. A manual differential counts 100 or 200 leukocytes, and the variability this implies has long been known [1]. Automated systems add a degree of freedom: they choose which fields to examine and when to stop. For questions that hinge on rare cells, such as the blast proportion relative to a classification threshold [2], [3], what matters is not classifier accuracy on the examined fields but what can be said about the fields that were not examined.

Stopping rules based on *information saturation* answer this question implicitly: they stop when further fields would add little information about the specimen as the selector's embedding represents it. InfoDPP-PAC [4] is a carefully constructed recent instance for whole-slide histopathology, and its authors state that its certificate bounds residual information in the selection objective, not diagnostic performance. That distinction is our starting point: saturation of information about an embedding does not imply that rare, decision-relevant evidence has been seen. We ask four questions. RQ1: does information-saturation stopping fail systematically when decision-changing evidence is rare? RQ2: why does certifying the *absence* of rare evidence require near-exhaustive examination? RQ3: how many cells must be examined to certify a threshold decision under adaptive sampling? RQ4: where do the assumptions of anytime-valid audit methods break on real data?

Our contributions are:

- a pre-registered evaluation showing that, in the tested settings, the stopping point of saturation stopping barely changed with rarity, that its halting signal decoupled from critical evidence as cells were pooled into fields, that two other saturation rules behave alike, and that on real marrow tiles it did worse than uniform sampling in low-blast specimens (RQ1);
- a minimax lower bound showing that near-exhaustive examination is required of *every* strategy without location information, and a proposition that ties a relevance-ranked search to the model's ROC curve (RQ2);
- an evaluation of design-based, anytime-valid certification of blast-proportion assertions [5], [6]: validity, cost from 500 to 4,000 cells, and decisions at fixed budgets (RQ3);
- an empirical examination of the assumptions behind these guarantees (RQ4): classifier error, acquisition shift on two external sources, spatial clustering measured in exhaustively annotated marrow regions, unequal field sizes (with two valid constructions) and verifier error measured from real reader labels.

This is a methodological study. Most specimens are composites of real single-cell images, and no result establishes clinical diagnostic validity.

II. Related Work

Rümke characterised the binomial and finite-population variability of manual differentials [1]; for absence of rare events, the "rule of three" bounds a rate after zero events [7], and Proposition 1 is its design-based finite-population counterpart. InfoDPP-PAC [4], a recent preprint, selects patches greedily under a blended information-gain and relevance objective and stops when the largest information gain among unselected patches falls below a concentration-corrected threshold. Its certificate rests on a stability assumption whose

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constant and target are assumed rather than derived; our results do not depend on it. Blast recognition itself is well studied [8]; our question concerns examination and inference.

Risk-limiting audits test assertions about finite populations with a guaranteed bound on confirming a false assertion [9]. ALPHA [5] turns assertions into bounded-mean tests through an *assorter* and builds a test supermartingale for sampling without replacement; betting confidence sequences [10] and Ville's inequality [11] provide the anytime validity. Shekhar et al. [6] extend this to weighted, adaptive sampling without replacement with side information; stratified extensions exist [12]. Prediction-powered inference [13], prediction-powered e-values [14] and active inference [15] combine predictions with a verified subset, mostly as fixed-sample statements about a superpopulation; the tests used here are anytime-valid statements about one finite specimen.

III. Setting and Theory

A. Estimands

A specimen has N fields; field i holds m_i cells, B_i of them blasts, $M = \sum_i m_i$, and $p = \sum_i B_i/M$. We distinguish *detection* (has a critical field, one with at least one blast, been examined?), *estimation* of p, and *threshold certification* ($p > \theta$?). We use $\theta = 0.10$, the blast percentage at which the International Consensus Classification (2022) separates myelodysplastic syndrome with excess blasts from MDS/AML [3], and $\theta = 0.20$, the WHO 5th-edition threshold for acute myeloid leukemia without defining genetic abnormalities [2]. Both count blast equivalents, which our labels do not; the thresholds are analysed separately and are inputs to a procedure, not validated here. An examiner selects fields sequentially, possibly using features x and past labels, and a selected field is *verified*: its cells' labels are revealed. All probabilities are over the examiner's randomisation for a fixed specimen, so the guarantees are finite-population and design-based.

B. Certifying absence (RQ2)

Let C be the set of k critical fields and M_n the event that none is among n examined fields; let $h(N,k,n) = \binom{N-k}{n}/\binom{N}{n}$.

Proposition 1 (Uniform sampling): Under sampling of n fields without replacement, $\mathbb{P}(M_n \mid C) = h(N,k,n)$; for $k = 1$ this is $1 - n/N$, so $\mathbb{P}(M_n) \le \alpha$ iff $n \ge \lceil (1-\alpha)N \rceil$.

Assumption 1 (No location information): Given x, every placement C with $|C| = k$ is possible.

Theorem 1 (Minimax lower bound): Under Assumption 1, every policy, adaptive or not, randomised or not, that examines at most n fields satisfies $\max_{|C|=k} \mathbb{P}_\pi(M_n \mid x, C) \ge h(N,k,n)$. Uniform sampling attains the bound; every deterministic policy has worst-case miss probability 1 if $n \le N-k$. The proof (Appendix) fixes the examiner's randomness: on the all-negative run the policy examines a fixed set, and averaging over a uniformly placed C gives the bound (Yao's principle [16]). We verified all statements exactly in rational arithmetic and, for $N \le 7$, by solving the search game as a linear program. If k is unknown, the worst case is $k = 1$, so certifying absence at level α needs $\lceil (1-\alpha)N \rceil$ fields (Fig. 1). If verification misses an examined critical field with probability β, absence cannot be certified below β even by a census.

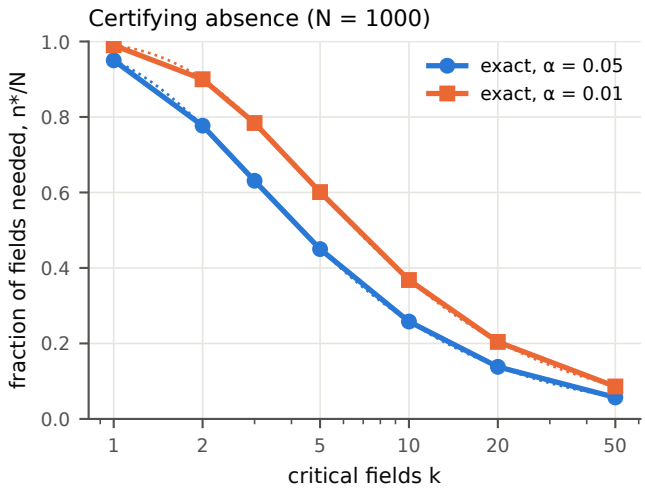


Fig. 1. Fraction of fields that any strategy without location information must examine to certify the absence of k critical fields ($N = 1000$; exact values solid, sufficient closed form $1 - \alpha^{1/k}$ dotted).

Informative features can let a search beat the bound on average, through a model of where evidence lies:

Proposition 2 (Ranked search): Let one field be critical with score $S_c \sim F_1$, and the other $N-1$ scores be independent with distribution F_0. A scan of the n highest-scoring fields misses the critical field with probability $\mathbb{E}_{S_c}[\mathbb{P}(\mathrm{Bin}(N-1, 1-F_0(S_c)) \ge n)]$, which tends to $1 - \mathrm{TPR}(t_r)$ as $N \to \infty$ with $n/(N-1) \to r$, where t_r is the threshold with false-positive rate r.

Given $S_c = s$, the critical field is missed iff at least n others score above s; the limit follows by dominated convergence. With $F_1 = F_0$ the formula returns $1 - n/N$. Certifying absence by ranking thus needs a sensitivity of about $1 - \alpha$ at a false-positive rate of n/N, a property of the model that no sampling design supplies. It is an average-case result under independence, neither a proof that only ranking helps nor a worst-case guarantee: a ranked scan surely misses a critical field placed among low-scoring fields.

C. Certifying a threshold assertion (RQ3)

Making "$p > \theta$" true requires more than θM blasts, which cannot hide in one field, so the bound above does not apply. With the assorter $a_j = b_j/(2\theta)$, $\bar{a} > 1/2 \iff p > \theta$, and the assertion becomes a one-sided bounded-mean test. Under uniform sampling of fields, ALPHA [5] gives a test supermartingale; rejecting when it exceeds $1/\alpha$ controls false certification at α at any data-dependent stopping time. For adaptive sampling, write $p = \sum_i w_i f_i$ with known weights and $f_i \in [0,1]$. If the field I_t is drawn from a predictable distribution q_t positive on all unsampled fields, the observation $Z_t = w_{I_t} f_{I_t}/q_t(I_t)$ has conditional mean equal to the unsampled mass, which under $H_0 : p \le \theta$ is at most

TABLE I
WHAT EACH GUARANTEE ASSUMES, AND WHAT HAPPENED WHEN THE ASSUMPTION WAS TESTED.

Assumption	Needed for	Test and result
No location information	Theorem 1	Partly fails on composites; ranking helps on average, through a model (Prop. 2)
Selection probabilities known, positive	Validity	Uncorrected ranking: false certification 0.89–0.99
Analysis unit = sampled unit	Validity	Cells as draws: up to 0.34 per real specimen, 0.33–0.48 under strong clustering
Target is cell-weighted p	Validity	Averaging proportions: 0.32–0.38; Prop. 3 restores ≤ 0.005
Verified label = truth	Validity for true p	Single reader: 0.28 ("above"), 0.63 ("below"); margin 0.03 sufficed for "above"
Classifier accurate, exchangeable errors	Efficiency only	Validity held under shift (≤ 0.014); saving reversed (1.77× cells)

$\mu_t = \theta - \sum_{s<t} w_{I_s} f_{I_s}$; a betting process $\prod_s (1+\lambda_s(Z_s-\mu_s))$ with predictable, non-negative, bounded bets is a test supermartingale whatever the classifier does [6]. We use their prop-MS sampler, $q_t(i) \propto w_i S_i$ with a floor (S_i the field's mean classifier score), and approximate-Kelly bets. Validity requires only known, positive q_t and verified labels.

Averaging field proportions, however, estimates $N^{-1}\sum_i B_i/m_i$, which is not p when field sizes differ.

Proposition 3 (Unequal field sizes): (a) If every m_i is known, $w_i = m_i/M$ and $f_i = B_i/m_i$ give $\sum_i w_i f_i = p$, so the weighted test is valid for $H_0 : p \leq \theta$. (b) If only $m_i \leq m_{\max}$ is known, $a_i = (B_i - \theta m_i + \theta m_{\max})/m_{\max} \in [0,1]$ and $N^{-1}\sum_i a_i > \theta \iff p > \theta$, so equal-weight tests on a_i with null mean θ are valid.
For (b), $0 \leq B_i \leq m_i \leq m_{\max}$ bounds a_i, and $\sum_i a_i = (\sum_i B_i - \theta M)/m_{\max} + N\theta$. The margin shrinks by $\bar{m}/m_{\max}$, so a loose bound costs power.

Remark 1 (Imperfect verification): If the verifier's false blast calls are at most β_0 times the true non-blasts, its proportion satisfies $\tilde{p} \leq p + \beta_0(1-p)$, so testing $\tilde{p} > \theta + \beta_0(1-\theta)$ is valid for $p > \theta$; symmetrically, testing $\tilde{p} < \theta(1-\beta_1)$ is valid for $p < \theta$ if missed blasts are at most β_1 times the true blasts. The bound concerns the verifier, not the design.

We compare, on the same specimen, ALPHA and betting with uniform sampling; betting with classifier-driven sampling, with and without a control variate; an invalid *uncorrected* design that ranks fields by score but analyses them as uniform draws; and conventional 100- and 200-cell counts. Table I lists what each guarantee assumes and where we tested it.

IV. EXPERIMENTAL DESIGN

A. Pre-registration and status of analyses

Hypotheses, decision rules, seeds and scripts were frozen in five git tags, each pushed before the corresponding data were generated; first confirmatory outputs followed the push by 43 s to 18 min (audit table in the repository). "Confirmatory" means fresh Monte Carlo draws on a disjoint seed family over the same cell pool, which protects against analytic flexibility but not against effects specific to that pool; only H4a–c and H7a used data not analysed before. Several hypotheses check the implementation or a theorem rather than an empirical claim (marked V in Table V). All later analyses used new seed families and are labelled exploratory; all deviations are logged in the repository.

B. Data

KU-Optofil [17] (CC BY 4.0): Sysmex DI-60 images of May–Grünwald–Giemsa peripheral blood smears labelled by haematologists; we used 28,280 cells from 275 identified patients (13 classes) in five patient-grouped folds. Blasts come from only 11 patients (489 blasts). *RV-PBS* [18]: 752 annotated cells (153 blasts) from phone-camera fields, used only as an acquisition-shift probe; no licence is stated. *MLL23* [19] (CC BY 4.0): 41,621 peripheral-blood cells (Pappenheim stain, Metafer scanner), 8,606 myeloblasts; we excluded atypical promyelocytes (blast equivalents our labels do not count), smudge cells and normoblasts, leaving 36,529; there are no patient identifiers. *BoneMarrowWSI* [20], [21] (CC BY 4.0): pediatric bone-marrow aspirate slides with 228 exhaustively annotated regions of about 225 μm; we used the annotations, the per-round reader labels, and the image pixels of the annotated regions.

C. Representation, teachers and specimens

Cells were embedded with a frozen DINOv2 ViT-S/14 [22] (224 px); a logistic probe cross-fitted by patient fold reached an out-of-fold AUC of 0.979. The pre-registered *cell-level teacher* scores a field by its largest cell score, which presupposes cell detection; an exploratory *field-level teacher* is a logistic regression on the pooled field embedding, the representation the selector sees. Part I composites have $N = 200$ fields of $m \in \{1, 5, 10, 20\}$ cells (mean-pooled embeddings) with $k \in \{1, 2, 5, 10, 20\}$ critical fields of one blast. Part III composites have 100 fields of 5 cells ($M = 500$) with p fixed exactly, blasts from all 489 blasts and background cells from blast-free patients; 10% of that background are platelets and erythroblasts, so p is the blast share of imaged objects (a leukocyte-only background changed nothing material). Exploratory extensions use M up to 4,000 and field sizes drawn from BoneMarrowWSI.

D. InfoDPP-PAC and baselines

No code has been released. We implemented the objective, greedy blended gain and halting rule (Eqs. 3–7 of [4]) with the published weights, set $\rho = 1$, and calibrated τ on blast-free specimens of other folds so that the median stop is 16.3% of all fields. The original runs stopped at 16.3% of a 300-patch budget, about 2.4% of a slide (a ratio of reported averages, not the original configuration), so our setting gives InfoDPP-PAC about seven times its own examination share; we also report a 2.4% run. A random seed of 5 fields fits the Gaussian process and is not counted as examined. Baselines were uniform sampling at the same per-episode cost (paired), a fixed field order, confidence-based stopping (stop at the first field with relevance ≥ 0.9 or after 30 consecutive fields ≤ 0.1), and exhaustive examination.

E. Statistics

Miss rates: Wilson intervals, exact McNemar tests, fold-clustered bootstrap intervals (pre-registered) and, because five folds allow only 126 bootstrap resamples, t intervals on fold estimates. False certification: Clopper–Pearson intervals and a one-sided exact binomial test of rate ≤ 0.05. Costs: paired Wilcoxon tests with Holm correction and ratios of medians with bootstrap intervals. Real-data analyses bootstrap over patients. $\alpha = 0.05$ unless stated. Our ALPHA and betting statistics agree with the reference implementation of ALPHA (SHANGRLA) to within 3×10^{-13} in log scale on 2,000 random cases; the only difference, a conservative rule missing from our frozen ALPHA, changed no median cost. A suite of 26 deterministic tests, with adversarial and adaptive samplers and negative controls, is in the repository.

V. Results

A. Information-saturation stopping (RQ1)

In 24,000 confirmatory episodes the median stop was 32–33 of 200 fields in every cell, varying by at most one field across rarity levels (H1a). This is an empirical finding in the tested settings: the halting statistic has no relevance term, but it depends on the embeddings and the selected set, which depend on the cells. The AUC with which the information-gain term separates critical from non-critical fields fell from 0.778 at $m = 1$ to 0.593, 0.552 and 0.525 at $m = 5, 10, 20$, whereas the field-level teacher on the same pooled embeddings kept 0.978, 0.925, 0.882 and 0.827 (Kill-test 1; the pre-registered cell-level teacher gave 0.977–0.896). The evidence survives pooling; the halting signal does not use it.

At $k = 1$, InfoDPP-PAC missed the critical field in 0.315, 0.681, 0.759 and 0.805 of specimens at $m = 1, 5, 10, 20$; uniform sampling of the same size missed 0.822, 0.830, 0.846 and 0.828, matching $1 - n/N$ (H1d; Fig. 2). The advantage shrank with dilution for every k (H1b), but the pre-registered claim of no advantage at $m \geq 10$ failed (H1c): the difference was -0.087 $[-0.107, -0.067]$ at $m = 10$ and -0.023 at $m = 20$, whose fold-level t interval $[-0.050, 0.003]$ is borderline. Dilution, not rarity, governed the gap; rarity drove the absolute miss rate of every method.

Three saturation rules, calibrated to the same cost on 4,000 new specimens, missed within 0.04 of each other in every cell: InfoDPP-PAC, pure information-gain sampling and k-center coverage stopping missed 0.31–0.34 at $m = 1$ and 0.83–0.85 at $m = 20$, against 0.84 and 0.83 for uniform sampling. A scan ranked by the field-level teacher missed 0.05, 0.14, 0.21 and 0.37; Proposition 2 predicted 0.051, 0.146, 0.221 and 0.394. In these experiments relevance, not the stopping criterion, separated the rules. At about 2.4%, InfoDPP-PAC made 4–5 selections after its seed and missed 0.81–0.97 against 0.97–0.98 for uniform sampling, a clear advantage only with single-cell fields, whether or not the seed is counted.

On real image tiles (BoneMarrowWSI; 3,040 tiles of about $56\,\mu$m from 99 patients, specimens of 16–32 tiles, 200 runs each), InfoDPP-PAC stopped after 5.4 tiles on average. In the 24 low-blast specimens (at most a quarter of tiles critical) it missed every critical tile in 0.80 of runs, against 0.70 for uniform sampling, the exact value of Proposition 1: a difference of $+0.105$ $[0.002, 0.203]$, worse in 18 of 24 specimens. Its failures were specimen-specific: it missed in at least 95% of runs in 15 specimens, as Theorem 1 predicts for a nearly deterministic selector, whereas uniform sampling stayed near $h(N, k, n)$ everywhere. The teacher's tile-level AUC was 0.966 overall but 0.69 within these specimens, so ranking missed 0.54.

B. Certified threshold decisions (RQ3)

In 12,000 confirmatory 500-cell specimens, all four valid designs kept false certification at $p = \theta$ between 0.001 and 0.022 (H3a), whereas the uncorrected design certified falsely in 0.993 ($\theta = 0.10$) and 0.889 ($\theta = 0.20$) of boundary specimens (H3b; Fig. 3). Classifier-driven betting needed fewer cells than uniform ALPHA at every $p > \theta$ (H3c), for example 75 vs. 165 at $p = 0.15$, $\theta = 0.10$ and 145 vs. 250 at $p = 0.25$, $\theta = 0.20$, and cost fell with the margin (H3d). Separating the two changes, the sampler alone saved 27–72% of cells (0.28–0.73 times), whereas the betting test alone needed 1.03–1.83 times the cells of ALPHA.

With specimens of up to 4,000 cells (exploratory; 17,600 specimens), validity was unchanged (false certification ≤ 0.040 at $\alpha = 0.05$, ≤ 0.007 at $\alpha = 0.01$), and cost in cells was governed mainly by the margin (Table II); its dependence on M diminished at margins of 0.05 or more but stayed large near the threshold. At a margin of 0.05 or more it was within 20% of Wald's approximation $\log(1/\alpha)/\mathrm{KL}(p\|\theta)$ [23]; within 0.02 of the threshold it grew towards about 1,400 ($\theta = 0.10$) and 2,500 cells ($\theta = 0.20$). The near-census we had seen at $M = 500$ was a small-specimen effect. All savings are in expert-verified cells; the whole specimen must still be imaged and scored. Lowering α to 0.01 multiplied the cost by 1.0–1.7; the claim that $\alpha = 0.01$ is unattainable concerns conformal-risk-control calibration (negative risk budget below 99 positive episodes), which design-based tests do not use.

At fixed budgets (exploratory; 2,000-cell specimens, cells sampled individually), we compared three-way decisions ("above", "below", "undetermined"; Table III). Within 0.02 of the threshold a 100-cell count was wrong in 18–37% of specimens. Valid procedures, two one-sided sequential ALPHA tests or exact hypergeometric tests at $\alpha/2$ each, were wrong in at most 0.7% of specimens off the threshold, but at those margins returned "undetermined" in 61–98% of specimens at budgets up to 500 cells; the fixed-n test was decisive more often, the price of anytime validity.

C. Where the assumptions break (RQ4)

Classifier error. On KU-Optofil, blast recall at 95% specificity varied across the 11 blast patients from 0.39 to 1.00, false-positive rates across blast-free patients were overdispersed 6.6-fold, and errors concentrated in immature granulocytes and lymphoid cells. Certification stayed valid with blasts drawn only from low-recall patients (≤ 0.007), at 2.3 times the cost.

Acquisition shift. Applied unchanged, the KU-trained probe's blast recall fell to 0.340 [0.250, 0.421] on RV-PBS

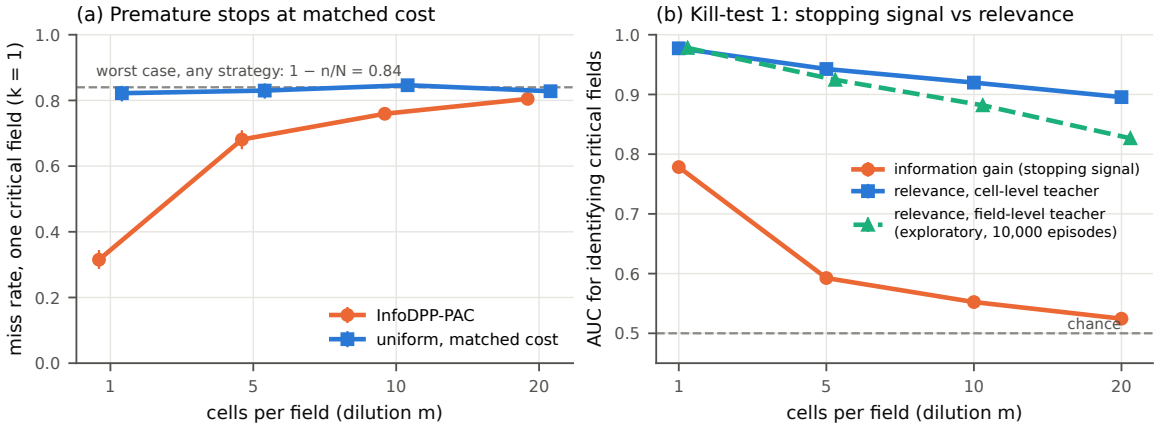


Fig. 2. RQ1, confirmatory run. (a) Miss rate with one critical field for InfoDPP-PAC and for uniform sampling of the same size (95% Wilson intervals); dashed: the worst-case bound $1 - n/N$. (b) AUC of the information-gain term and of the cell-level teacher for ranking critical above non-critical fields; dashed, exploratory: the field-level teacher on the same pooled embeddings as the selector (10,000 separate episodes).

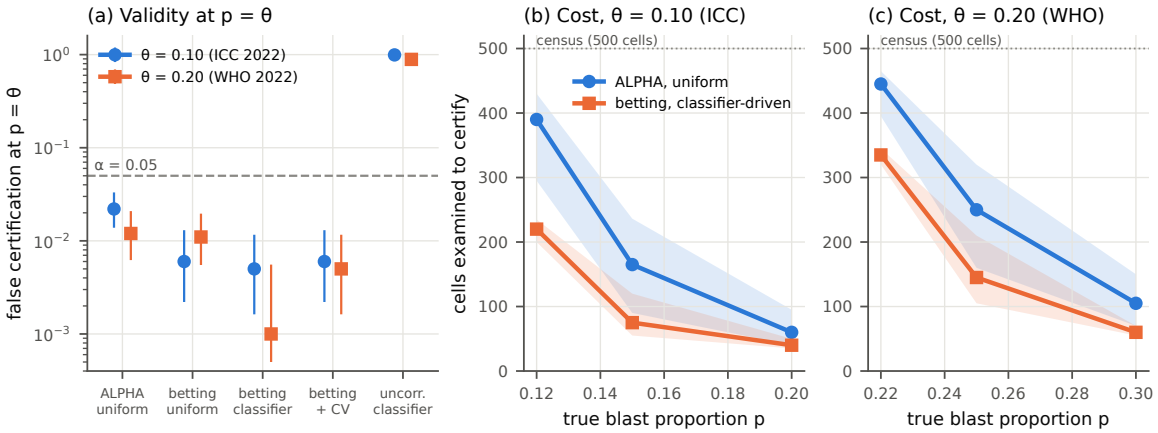


Fig. 3. RQ3, confirmatory run (500-cell specimens; $\alpha = 0.05$). (a) False certification at $p = \theta$ with 95% Clopper–Pearson intervals (log scale); CV: control variate; the uncorrected design analyses classifier-ranked fields as uniform draws. (b, c) Median cells to certification with interquartile bands.

TABLE II
MEDIAN CELLS TO CERTIFY $p > \theta$ AGAINST SPECIMEN SIZE M (EXPLORATORY; 400 SPECIMENS PER CELL). U: UNIFORM ALPHA; C: CLASSIFIER-DRIVEN BETTING; APPROX.: $\log(1/\alpha)/\mathrm{KL}(p\|\theta)$. †BLAST IMAGES REUSED, WHICH AFFECTS ONLY C.

		$M = 500$		$M = 1{,}000$		$M = 4{,}000$		
θ	p	U	C	U	C	U	C	Approx.
0.10	0.12	390	220	655	420	1,583	798	1,425
0.10	0.15	165	80	205	80	245†	85†	245
0.20	0.22	435	335	810	660	2,033†	2,043†	2,455
0.20	0.25	250	145	303	175	420†	185†	406

TABLE III
DECISIONS AFTER n INDIVIDUALLY SAMPLED CELLS (EXPLORATORY; $M = 2{,}000$; 2,000 SPECIMENS PER ROW). CONV.: SHARE WRONG FOR THE 100-CELL COUNT; EXACT AND SEQ.: SHARE CORRECT / WRONG FOR VALID FIXED-n AND SEQUENTIAL TESTS; THE REST IS "UNDETERMINED".

		$n = 100$			$n = 500$	
θ	p	Conv.	Exact	Seq.	Exact	Seq.
0.10	0.08	0.18	0.08 / <0.01	0.04 / <0.01	0.39 / 0	0.15 / <0.01
0.10	0.12	0.33	0.08 / <0.01	0.02 / <0.01	0.34 / 0	0.15 / <0.01
0.10	0.15	0.10	0.30 / <0.01	0.12 / <0.01	0.96 / 0	0.86 / <0.01
0.20	0.18	0.24	0.07 / <0.01	0.02 / <0.01	0.23 / 0	0.09 / <0.01
0.20	0.22	0.37	0.05 / 0.01	0.02 / <0.01	0.25 / <0.01	0.08 / <0.01
0.20	0.25	0.14	0.18 / 0	0.06 / 0	0.87 / 0	0.59 / 0

(H4a) and to 0.63 [0.62, 0.64] on MLL23 (AUC 0.786 and 0.915); its false-positive rate barely rose (0.058 and 0.060 against 0.050; H4b not supported). Every design stayed valid (RV-PBS 3 false certifications in 1,000, H4c; MLL23 0.001–0.011). Efficiency did not: classifier-driven sampling needed 0.28–0.73 times the cells of uniform sampling in distribution, 0.76 [0.70, 0.81] and 0.90 [0.82, 1.05] times on MLL23 ($p = 0.15$, 0.20), and 1.77 times on RV-PBS (195 vs. 110 cells at $p = 0.20$; H6d). A 50% uniform mixture stayed valid (H6a), kept 58–73% of the in-distribution saving (H6c) and cut the RV-PBS excess to +23% [11, 30], missing the pre-registered 20% bound (H6b). MLL23 and RV-PBS lack patient identifiers, so their intervals are too narrow; the three sources differ in more than recall.

Spatial clustering and the unit of analysis. Under strong

synthetic clustering (18,000 specimens), field-level designs stayed valid (≤ 0.042, H5a), whereas analysing the cells of sampled fields as independent draws certified falsely in 0.327 ($\theta = 0.10$) and 0.478 ($\theta = 0.20$) of boundary specimens (H5b); clustering doubled the cost of uniform field sampling (H5c). In 116 exhaustively annotated marrow regions of 75 patients, blasts were clustered (H7a; permutation $p = 0.024$ at the quadrant and 0.0005 at the 4×4 scale), but weakly: the excess intraclass correlation over its permutation null was 0.013 [0.002, 0.028] and 0.043 [0.018, 0.067] (exploratory bias correction). Using each patient's regions as a real specimen with 4×4 subregions as fields and the specimen's own proportion as threshold (an exact null; 400 sampling orders per specimen; exploratory), the weighted test of Proposition 3(a) certified falsely in 0.001 of runs and at most 0.01 in any specimen. The cell-level analysis gave 0.051 overall, close to α, yet exceeded 0.05 in 27 of 75 specimens (21 significantly), up to 0.335: validity is a statement about each specimen, and a pooled rate can hide failures. These specimens are small (median 105 cells) and mostly high-blast.

Unequal field sizes. Marrow quadrants hold 12–29 nucleated cells (interquartile range). With these sizes, averaging field proportions certified falsely in 0.38 [0.35, 0.41] and 0.32 [0.29, 0.35] of boundary specimens even with blasts placed independently of size, and in every specimen with blasts enriched in small fields: the test is valid for the mean of proportions, which is the wrong quantity. Both constructions of Proposition 3 stayed at or below 0.005; with known counts, classifier-driven sampling certified $p = 0.15$ against $\theta = 0.10$ with 297 cells against 646 for uniform sampling, and the bounded assorter needed 631 because $\bar{m}/m_{\max} \approx 0.24$.

Verifier error. BoneMarrowWSI publishes the label of each reading round; which person read which cell is not recorded [20]. The second reader called a blast in 1.1% [0.7, 1.7] of cells the first called a counted non-blast and a non-blast in 1.2% [0.8, 1.6] of the first reader's blasts; in a further 2.7% of cells one reader did not count the cell. With the first reader as verifier, supplying the labels and the field counts and weights, and the consensus as truth (83 real specimens, exact nulls), the reader's proportion differed from the truth by a median of 0.005 (maximum 0.075), and the weighted test certified "$p > \theta_s$" falsely in 0.28 and "$p < \theta_s$" in 0.63 of runs. Remark 1 with the readers' disagreement rates was not enough (0.16, 0.42); in this sample a margin of 0.03 sufficed for "above" but "below" still failed in three specimens at 0.05. These margins were selected in-sample and are direction-specific, not a validated correction. The conclusion is that a certificate verified by one reader certifies that reader's count; inference about the consensus proportion requires a separately established bound on verifier error.

One benchmark. Table IV puts these designs and conditions side by side. Where their assumptions held, the design-based tests kept false certification at or below 0.031; with a single reader as verifier, the test certified the reader's count. Every analysis without a design-based guarantee failed in some condition, although cells of fields as draws looked acceptable in distribution (0.041) and pooled over real arrangements (0.051). Validity transferred across conditions; efficiency did not.

D. Robustness

The pre-registered Part I sensitivity analyses (max pooling, a single-cell representation, operating points of 5–50% and a single-host background) changed no direction: the difference from uniform sampling at $m \geq 10$ never exceeded 0.13, and InfoDPP-PAC still missed 73–77% of single-critical-field specimens with max pooling. The single-host background enlarged its residual advantage (-0.27 at $m = 10$, -0.15 at $m = 20$), so "the advantage vanishes by $m = 20$" depends on the background construction. For Part III, confusable and leukocyte-only backgrounds left every design valid (Table IV), and the default ALPHA prior guess was the cheapest of three fixed guesses. Table V lists all pre-registered verdicts; three of 20 hypotheses were not supported.

VI. Discussion

Our results separate three quantities that are often reported together. *Model confidence* about examined fields says nothing about unexamined ones. *Information saturation* concerns the selector's representation of the specimen; its association with decision-relevant evidence disappeared once cells were pooled into fields, although the pooled representation still carried the evidence, and on real tiles saturation-based selection did worse than uniform sampling. *Design-based certification* bounds the probability of a false threshold decision for the specific specimen, provided selection probabilities are known and positive and labels come from verification. Only the third is a guarantee about what was not examined. It held in every condition we tested, specimen by specimen, including a classifier that lost two thirds of its blast recall; the classifier affected only how quickly certification was reached.

Four practical consequences follow. First, certify a proportion against a stated threshold rather than the absence of rare cells, which requires a near-census; its cost in verified cells is governed mainly by the margin, and depends on specimen size only weakly once specimens are large and the margin is not small. Second, record selection probabilities, analyse at the unit that was sampled, and weight fields by their cell counts. Third, allow and report "undetermined": near the threshold it is the honest outcome of a few hundred cells, and a count that forces a call is wrong in up to a third of specimens. Fourth, treat verifier error as part of the claim: a certificate concerns the verifier's count, and relating it to the true proportion needs a margin that bounds verifier error, established on separate data. A procedure that is valid but usually undetermined near the threshold is statistically sound and operationally limited; making it decisive requires more cells, not a more confident model.

Limitations. Most specimens are composites from one laboratory, and all KU-Optofil blasts come from 11 patients, so cost and miss-rate comparisons are imprecise. The real-data spatial, tile and verifier analyses use pediatric bone-marrow aspirates, not peripheral blood, so they validate the method but not clinical use, and the specimen-level validity test uses each specimen's own proportion, not fixed diagnostic thresholds;

TABLE IV

ONE BENCHMARK ACROSS CONDITIONS (STORED RESULTS; NO NEW SIMULATION). TOP: SHARE OF SPECIMENS CERTIFYING "$p > \theta$" AT $p = \theta$ ($\alpha = 0.05$; LARGER OF $\theta = 0.10, 0.20$; 400–1,000 SPECIMENS); BONEMARROWWSI: POOLED OVER RUNS WITH $\theta_s = p_s$ [LARGEST SPECIMEN RATE]. BOTTOM: MEDIAN CELLS TO CERTIFY $p = 0.15$ AGAINST $\theta = 0.10$ (UNCERTIFIED SPECIMENS COUNT ALL CELLS). SPECIMENS: 500 CELLS; ABOUT 1,300 WITH UNEQUAL FIELDS; MEDIAN 105 REAL CELLS. STRESSED: ASSUMPTION TESTED (UNIT: ANALYSED UNIT = SAMPLED UNIT; TARGET: CELL-WEIGHTED p). C/X: CONFIRMATORY/EXPLORATORY RUN. [a]UNEQUAL FIELDS AND BONEMARROWWSI: ALPHA ON THE BOUNDED ASSORTER, BETTING WITH KNOWN COUNTS (PROP. 3). [b]COINCIDES WITH ALPHA FOR EQUAL FIELDS; 1.00 WITH BLASTS ENRICHED IN SMALL FIELDS. ONE READER WITH MARGIN 0.03: 0.0003 [0.013].

	Composites of KU-Optofil cells				External cells		BoneMarrowWSI	
	In distr.	Clustered	Confusable backgr.	Unequal fields	RV-PBS	MLL23	Real arrangem.	One reader
Stressed	—	*unit*	*classifier*	*target*	*classifier*	*transfer*	*geometry*	*verifier*
Status	C	C	X	X	C	X	X	X
False certification of "$p > \theta$" at $p = \theta$								
ALPHA, uniform[a]	0.022	0.025	0.031	0	—	—	0 [0]	—
Betting, uniform[a]	0.011	0.005	0.011	0.005	0.011	0.011	0.001 [0.010]	0.277 [1.00]
Betting, classifier-driven	0.005	0.003	0.007	0.001	0.001	0.001	—	—
Betting, 50% mixture	0.014	—	—	—	0.003	0.004	—	—
Ranked, analysed as uniform	0.993	—	—	—	—	—	—	—
Cells of fields as draws	0.041	0.478	—	—	—	—	0.051 [0.335]	—
Mean of field proportions[b]	—	—	—	0.379	—	—	0.062 [1.00]	—
100-cell count, "above"	0.427	0.388	—	—	—	—	—	—
Median cells to certify $p = 0.15$ against $\theta = 0.10$								
ALPHA, uniform[a]	165	345	160	1,031	—	—	—	—
Betting, uniform[a]	270	355	265	646	260	283	—	—
Betting, classifier-driven	75	55	105	297	265	215	—	—
Betting, 50% mixture	145	—	—	—	285	220	—	—

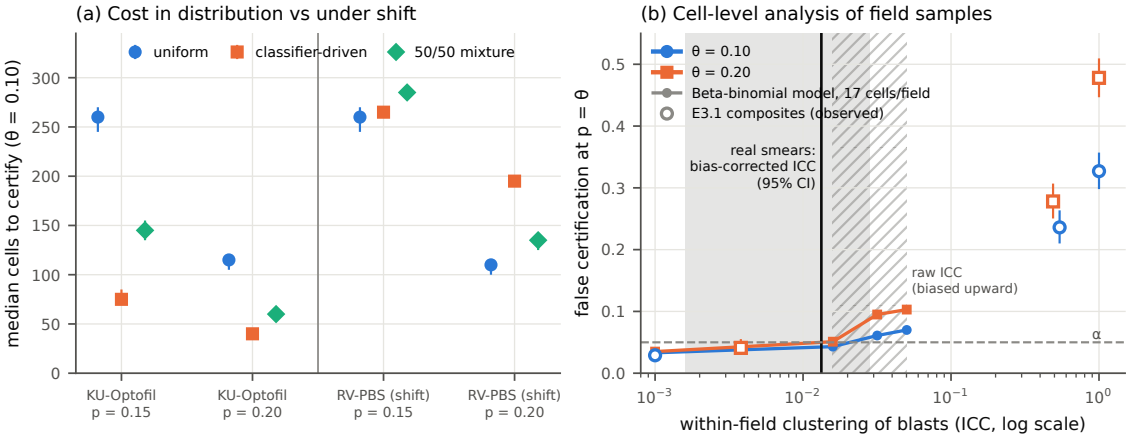


Fig. 4. RQ4. (a) Median cells to certify $p > 0.10$ in distribution (KU-Optofil) and under acquisition shift (RV-PBS); all designs keep false certification ≤ 0.014. (b) False certification at $p = \theta$ when cells of sampled fields are analysed as individual draws: composites with scattered or hot-field arrangements (markers) and a Beta-binomial model (lines). Solid band and line: bias-corrected clustering of real marrow regions (exploratory); hatched: the raw, upward-biased estimate, for comparison.

their specimens are small and mostly high-blast, and specimens near the diagnostic thresholds are barely represented. The external sources lack patient identifiers. Reading rounds are not identified experts, and the verifier margins were chosen in-sample. Many analyses were added after the pre-registered ones and are exploratory. InfoDPP-PAC's unreported hyperparameters were calibrated to a cost, though two other saturation rules gave the same result. We used one frozen encoder and did not implement prediction-powered e-values [14].

VII. CONCLUSION

Information saturation does not certify what remains unexamined, and without location information certifying the absence of rare evidence requires examining nearly the whole specimen. Threshold decisions about the blast proportion can be certified with finite-sample, anytime-valid guarantees, whose validity rests on the sampling design, the unit of analysis and the verifier rather than on the classifier. Fully annotated peripheral-blood smears with patient identifiers, and verification by identified haematologists, are the most important next steps.

DATA AND CODE AVAILABILITY

Code, the pre-registration with its git history and audit table, the deviation log and the analysis outputs are available at https://github.com/Afridi-7/certified-adaptive-pbs (to be archived

TABLE V
PRE-REGISTERED HYPOTHESES (E: EMPIRICAL CLAIM; V: IMPLEMENTATION OR THEOREM CHECK).

ID		Hypothesis (abridged)	Verdict
H1a	E	Stop point unchanged across rarity levels	yes
H1b	E	Advantage shrinks with dilution	yes (4/4)
H1c	E	No advantage (± 0.05) at $m \geq 10$	**no**
H1d	V	Uniform miss rate $= 1 - n/N$	yes
KT1	E	Information gain and relevance separable	yes
H3a	V	Four designs valid at both thresholds	yes
H3b	E	Uncorrected adaptive selection invalid	yes
H3c–d	E	Classifier-driven cheaper; cost falls with margin	yes
H4a	E	Blast recall drops under shift	yes
H4b	E	False-positive rate rises under shift	**no**
H4c	V	Certificate valid with shifted scores	yes
H5a	V	Field-level tests valid under clustering	yes
H5b–c	E	Cell-level analysis invalid; clustering raises cost	yes
H6a	V	Defensive mixture valid	yes
H6b	E	Mixture caps loss under shift below 20%	**no**
H6c–d	E	Mixture keeps $\geq$50% saving; reversal replicates	yes
H7a	E	Blasts clustered in real smears	yes

on Zenodo with a DOI). KU-Optofil, BoneMarrowWSI-PediatricLeukemia and MLL23 are public under CC BY 4.0; RV-PBS is distributed by its authors without a stated licence and was used for research only.

ACKNOWLEDGMENT

The author thanks Dr. Balázs Harangi for supervising this work.

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APPENDIX
PROOF OF THEOREM 1

Fix x and a realisation u of the examiner's randomness. On the run in which every examined field is negative, every input to the policy is determined by (x, u), so the run examines a fixed set $A(u)$ with $|A(u)| \leq n$. If $C \cap A(u) = \emptyset$, the actual run follows this path and misses C; otherwise it follows the path until the first element of C in $A(u)$, which it examines. Hence $\mathbb{P}_\pi(M_n \mid x, C) = \mathbb{P}_U(C \cap A(U) = \emptyset)$. Let C be uniform over k-subsets, independent of U, which Assumption 1 permits. For fixed u, the fraction of k-subsets disjoint from $A(u)$ is at least $\binom{N-n}{k} / \binom{N}{k} = h(N, k, n)$. Averaging over U and noting that the maximum over C is at least the average gives the bound; uniform sampling attains it by Proposition 1. For a deterministic policy A is one set with $|A| \leq N - k$, and placing C outside it gives miss probability 1.